DGMS 2019 • SECTION G: SEISMOLOGY & EARTH INTERIOR / CHEMICAL OCEANOGRAPHY

Question 65: Relative Abundance of Dissolved Salts in Ocean Water

Dittmar's Principle (1884), Major Ionic Stoichiometry, and Evaporite Mineral Crystallization

OFFICIAL EXAMINATION QUESTION

Which one of the following is the correct order of abundance of dissolved salts in ocean water?

(a) $NaCl > MgSO_4 > MgCl_2 > CaCO_3 > K_2SO_4 > CaSO_4 > MgBr_2$

(b) **$NaCl > MgCl_2 > MgSO_4 > CaSO_4 > K_2SO_4 > CaCO_3 > MgBr_2$ [CORRECT]**

(c) $NaCl > K_2SO_4 > MgSO_4 > MgCl_2 > CaCO_3 > CaSO_4 > MgBr_2$

(d) $NaCl > MgCl_2 > K_2SO_4 > CaSO_4 > CaCO_3 > MgBr_2 > MgSO_4$

VERIFIED ANSWER KEY

Correct Option: (b) $NaCl > MgCl_2 > MgSO_4 > CaSO_4 > K_2SO_4 > CaCO_3 > MgBr_2$ — Based on William Dittmar's classical stoichiometric analysis of 77 ocean water samples from the *HMS Challenger* expedition (1884), the major dissolved salts in normal seawater (salinity 35‰) follow this exact hierarchy: Sodium chloride (77.76%), Magnesium chloride (10.88%), Magnesium sulfate (4.74%), Calcium sulfate (3.60%), Potassium sulfate (2.46%), Calcium carbonate (0.34%), and Magnesium bromide (0.22%).

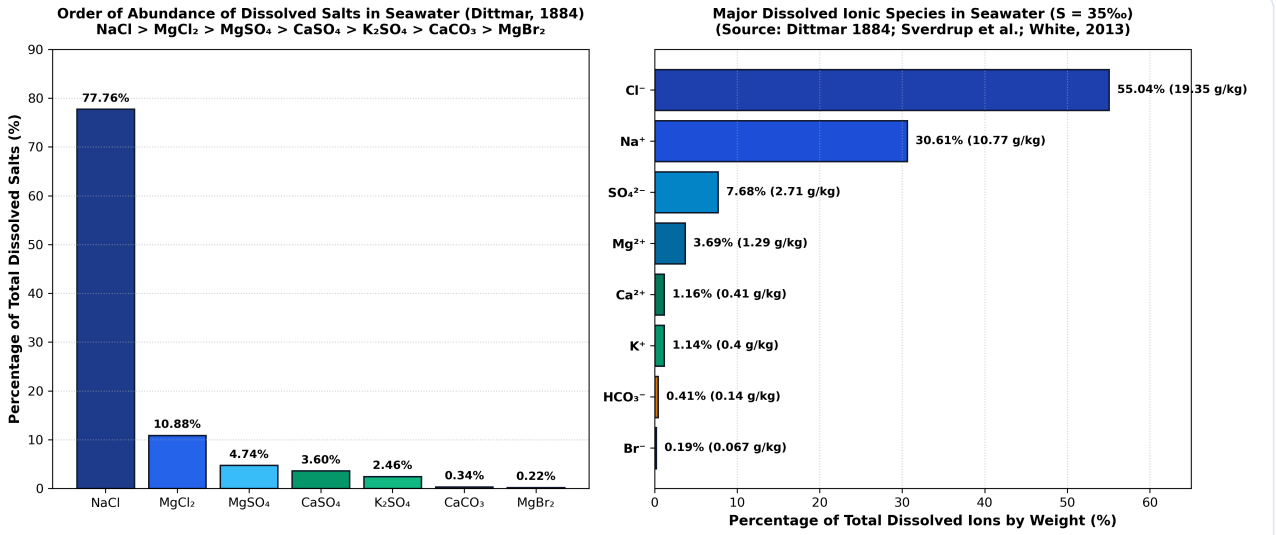


Figure 65.1 | Stoichiometric Abundance of Dissolved Salts and Major Ions in Seawater: (Left) Quantitative percentage distribution of dissolved salts in seawater according to the classic Dittmar (1884) formulation. NaCl dominates overwhelmingly (~78%), followed by $\mathrm{MgCl_2}$ (~11%) and $\mathrm{MgSO_4}$ (~4.7%). (Right) Relative weight abundance of individual dissolved ions in standard ocean water of 35‰ salinity (White, 2013; Sverdrup et al., 1942).

1. Chemical Oceanography & Dittmar's Principle of Constant Proportions

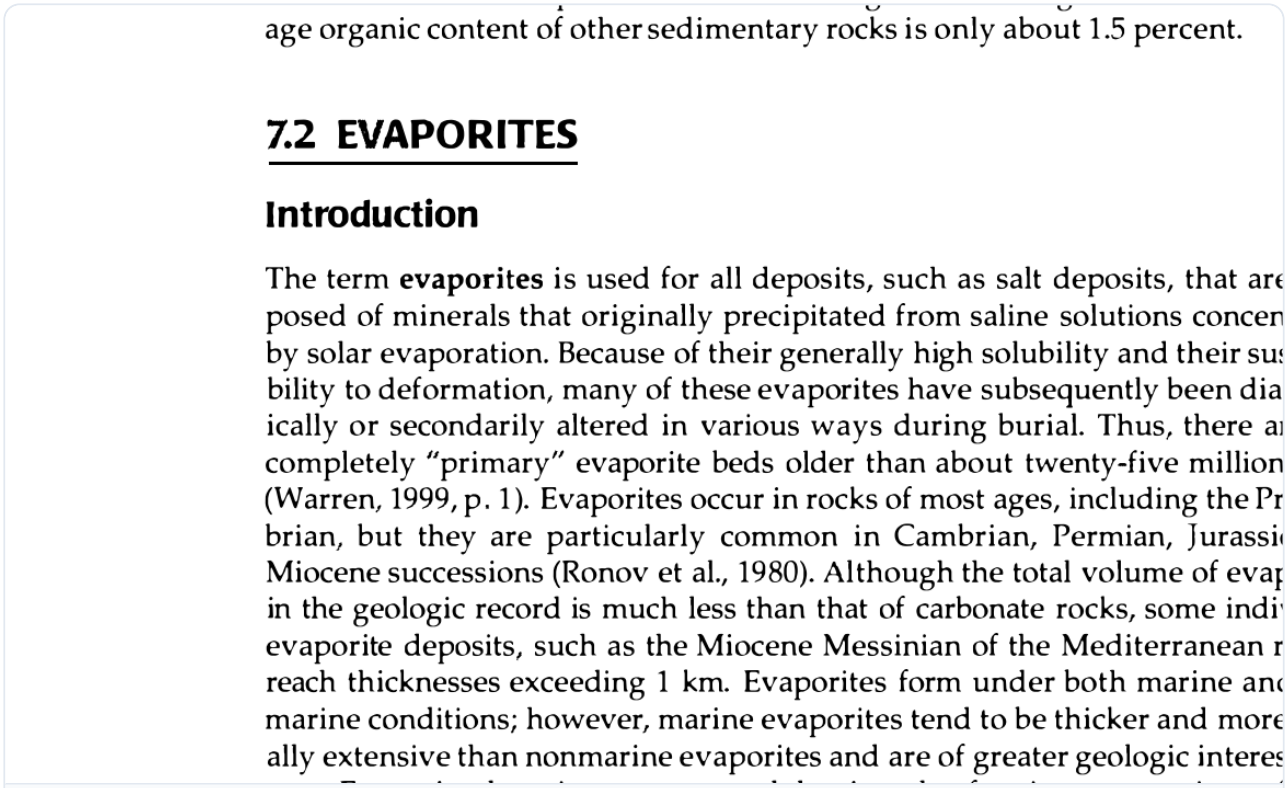
Although seawater contains virtually every natural element in trace amounts, over 99.9% of all dissolved material consists of just eleven major conservative constituents. In 1884, Scottish chemist **William Dittmar** analyzed seawater samples collected across all global oceans during the historic *HMS Challenger* expedition (1872–1876).

Dittmar established what is known as **Marcet's Principle** or **Dittmar's Principle of Constant Proportions**: *regardless of variations in total salinity from place to place (typically 33‰ to 38‰ in the open ocean), the ratios of the major dissolved ions to one another remain virtually constant across the global oceans*.

2. Quantitative Breakdown of Dissolved Salts in Seawater (S = 35‰)

When the individual dissolved cations ($\mathrm{Na^+, Mg^{2+}, Ca^{2+}, K^+}$) and anions ($\mathrm{Cl^-, SO_4^{2-}, HCO_3^-, Br^-}$) are combined into hypothetical neutral salts (as conventionally reported in classical chemical oceanography and geology textbooks like P. K. Mukherjee, G. B. Mahapatra, and K. M. Bangar), they yield the following standard composition:

Rank	Dissolved Salt	Chemical Formula	Concentration in 1 kg Seawater	Weight Percentage of Total Salt
1	**Sodium chloride**	NaCl	**27.213 g**	**77.76%**
2	**Magnesium chloride**	$\mathrm{MgCl_2}$	**3.807 g**	**10.88%**
3	**Magnesium sulfate**	$ \mathrm{MgSO_4}$	**1.658 g**	**4.74%**
4	**Calcium sulfate**	$\mathrm{CaSO_4}$	**1.260 g**	**3.60%**
5	**Potassium sulfate**	$ \mathrm{K_2SO_4}$	**0.863 g**	**2.46%**
6	**Calcium carbonate**	$\mathrm{CaCO_3}$	**0.123 g**	**0.34%**
7	**Magnesium bromide**	$\mathrm{MgBr_2}$	**0.076 g**	**0.22%**
Total Dissolved Solids (Salinity)			**35.000 g / kg**	**100.00%**



age organic content of other sedimentary rocks is only about 1.5 percent.

7.2 EVAPORITES

Introduction

The term **evaporites** is used for all deposits, such as salt deposits, that are posed of minerals that originally precipitated from saline solutions concen by solar evaporation. Because of their generally high solubility and their sus bility to deformation, many of these evaporites have subsequently been dia ically or secondarily altered in various ways during burial. Thus, there ar completely "primary" evaporite beds older than about twenty-five million (Warren, 1999, p. 1). Evaporites occur in rocks of most ages, including the Pr brian, but they are particularly common in Cambrian, Permian, Jurassi Miocene successions (Ronov et al., 1980). Although the total volume of evap in the geologic record is much less than that of carbonate rocks, some indi evaporite deposits, such as the Miocene Messinian of the Mediterranean r reach thicknesses exceeding 1 km. Evaporites form under both marine and marine conditions; however, marine evaporites tend to be thicker and more ally extensive than nonmarine evaporites and are of greater geologic interes

Figure 65.2 | Evaporite Mineral Classes and Marine Salt Precursors: Mineralogical classification of marine evaporite salts into Chlorides (Halite, Sylvite, Carnallite), Sulfates (Anhydrite, Gypsum, Kieserite, Polyhalite), and Carbonates (Calcite, Dolomite, Magnesite). (Source: Sam Boggs Jr., *Principles of Sedimentology and Stratigraphy*, 4th Ed., Table 7.1, p. 199).

3. Usiglio's Evaporation Sequence Connection

The relative abundance of dissolved salts directly governs the progressive precipitation sequence observed during solar evaporation of marine water (Usiglio, 1849; Boggs, 2006):

- **Volume reduced to ~50% (Salinity ~70‰):** Trace **$\mathrm{CaCO_3}$** (calcite/aragonite) precipitates first because of its very low solubility product (K_{sp}).
- **Volume reduced to ~19% (Salinity ~180‰): $\mathrm{CaSO_4 \cdot 2H_2O}$** (gypsum) begins massive precipitation.
- **Volume reduced to ~9.5% (Salinity ~360‰): NaCl** (halite) crystallizes in massive beds, reflecting its 77.8% abundance.
- **Volume reduced to <5%:** Highly soluble magnesium and potassium "bittern salts" precipitate: **$\mathrm{MgSO_4}$** (epsomite/kieserite), **$\mathrm{MgCl_2}$**, **KCl** (sylvite), and carnallite ($\mathrm{KMgCl_3 \cdot 6H_2O}$).

4. Comprehensive Option Analysis

- **(a) $NaCl > MgSO_4 > MgCl_2 > CaCO_3 > K_2SO_4 > CaSO_4 > MgBr_2$:** *Incorrect.* Inverts $\mathrm{MgSO_4}$ and $\mathrm{MgCl_2}$ ($\mathrm{MgCl_2}$ is 10.9% while $

\mathrm{MgSO_4}$ is only 4.7%), and places $\mathrm{CaCO_3}$ ahead of $\mathrm{CaSO_4}$.

- **(b) NaCl > $MgCl_2$ > $MgSO_4$ > $CaSO_4$ > K_2SO_4 > $CaCO_3$ > $MgBr_2$: Correct.** Exact sequence matching Dittmar's classic table: $\mathrm{NaCl}~(77.8\%) > \mathrm{MgCl_2}~(10.9\%) > \mathrm{MgSO_4}~(4.7\%) > \mathrm{CaSO_4}~(3.6\%) > \mathrm{K_2SO_4}~(2.5\%) > \mathrm{CaCO_3}~(0.3\%) > \mathrm{MgBr_2}~(0.2\%)$.
- **(c) NaCl > K_2SO_4 > $MgSO_4$ > $MgCl_2$ > $CaCO_3$ > $CaSO_4$ > $MgBr_2$:** *Incorrect.* Grossly overestimates potassium sulfate (K^+ is minor compared to Mg^{2+}).
- **(d) NaCl > $MgCl_2$ > K_2SO_4 > $CaSO_4$ > $CaCO_3$ > $MgBr_2$ > $MgSO_4$:** *Incorrect.* Wrongly places $\mathrm{MgSO_4}$ at the very bottom, even though it is the third most abundant salt.

5. Authoritative Textbook References

- **William M. White (2013)** *Geochemistry*, Wiley-Blackwell, Chapter 15 ("The Oceans as a Chemical System"), Table 15.1 ("Composition of Standard Seawater"), pp. 645–665.
- **Sam Boggs Jr. (2006)** *Principles of Sedimentology and Stratigraphy*, 4th Ed., Pearson / Prentice Hall, Chapter 7 ("Evaporites"), Table 7.1, pp. 198–205.
- **P. K. Mukherjee (2013)** *A Text Book of Geology*, World Press, Chapter on Oceanic Deposits and Seawater Salinity.
- **H. U. Sverdrup, M. W. Johnson & R. H. Fleming (1942)** *The Oceans: Their Physics, Chemistry, and General Biology*, Prentice-Hall, Chapter 6 ("Chemistry of the Sea"), pp. 165–227.